
Skincare ingredients recommended by cosmetic dermatologists: A Delphi consensus study



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Background: There is ambiguity regarding the topical cosmetic ingredients preferred for common skin complaints.

Objective: To determine which topical ingredients are frequently recommended by cosmetic dermatologists for fine lines and wrinkles, acne, redness, dark spots, large pores, dry skin, and oily skin.

Methods: Literature review to develop long list of ingredients. Reduced by expert panel to most salient ingredients. Two rounds of Delphi consensus survey with second expert panel of clinicians and teachers. Comparative literature review to summarize published evidence supporting each consensus ingredient.

Results: List of 318 ingredients reduced by a panel of 17 dermatologists to 83. Two Delphi rounds completed by 62 dermatologists at 43 centers. Consensus achieved for 23 ingredients, including the

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following: azelaic acid (acne, dark spots); benzoyl peroxide (acne, oily skin); glycolic acid (acne, dark spots); mineral sunscreen (fine lines and wrinkles, redness); niacinamide (redness, dark spots); retinoids (fine lines and wrinkles, acne, dark spots, large pores, oily skin); salicylic acid (acne, oily skin); vitamin C (fine lines and wrinkles, dark spots). Most consensus ingredients supported by level 1b or 2b evidence.

Limitations: Some ingredients based on expert opinion.

Conclusion: Consensus exists among expert cosmetic dermatologists regarding ingredients most useful for common dermatologic concerns. (J Am Acad Dermatol 2025;93:1509-25.)

Key words: consensus; cosmetic; Delphi; dermatologist; ingredients; recommended; skincare; study.

INTRODUCTION

The armamentarium of topical skincare ingredients has grown considerably in the past decade. Consumers and dermatologists now have access to numerous active ingredients intended for various specific indications. At present, however, there is little clarity regarding which ingredients are most recommended by experts in the field of cosmetic dermatology. Patients and physicians may feel overwhelmed by the many products available when selecting a skin care routine.^{1,2} This study aimed to determine which generic topical ingredients are most frequently recommended by experts in cosmetic dermatology for common dermatologic skin concerns.

METHODS

Study design

Qualitative research design based on a literature search informed by expert opinion and followed by Delphi process-based consensus survey.

Development of initial ingredient lists

A literature search (Pubmed, 1990-2020) was performed to identify discrete, generic cosmetic ingredients in use in the United States within both over-the-counter and prescription products, and intended for the treatment of 7 common cosmetic skin complaints: fine lines and wrinkles; acne; redness; dark spots; large pores; dry skin; and oily skin. Long lists of such ingredients were then

CAPSULE SUMMARY

- Consensus exists among cosmetic dermatologists regarding the topical ingredients recommended to treat common skin concerns. Most recommended ingredients are supported by published data.
- These consensus recommendations may provide a starting point for building a skincare routine and may increase consistency and clarity in communications between clinicians and patients.

reviewed by expert panel of cosmetic dermatologists who met the following criteria: all were American Board of Dermatology board-certified dermatologists; and all were either program directors of American Society for Dermatologic Surgery accredited Cosmetic Dermatology Surgery Fellowship programs or otherwise identified as internationally recognized experts in cosmetic dermatology. For each of the 7 skin concerns, panelists were asked to identify

the ingredients that they believed were most useful. Specifically, they were asked to indicate at least 3 ingredients (or associated products containing these ingredients) they routinely prescribed or recommended for each of the 7 cosmetic concerns. Ingredients that were listed by 2 or more panelists were included in the subsequent Delphi process.

Delphi consensus process

The ingredient lists were then critically evaluated by a separate panel of experts and refined through 2 rounds of Delphi survey and a virtual consensus meeting. Members of the Association of Academic Cosmetic Dermatology were invited to be Delphi panelists, and the Association of Academic Cosmetic Dermatology Patient Experience Committee served as the steering committee overseeing the consensus process.* Delphi panelists and steering committee members were: (1) American Board of Dermatology board-certified dermatologists; and (2) directors or co-directors of cosmetic and laser education at

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Accreditation Council for Graduate Medical Education accredited dermatology residency programs.

In the first survey round, Delphi panelists were asked to rate how likely they were to recommend each ingredient on a 9-point Likert scale, with 1 representing “strongly discourage use,” and 9 representing “strongly recommend use.” Panelists were also able to provide feedback regarding their opinions about each ingredient. Following the first Delphi round, the steering committee held a virtual consensus meeting. In this meeting, investigators (G.V.A., B.Y.K., and A.M.R.) facilitated discussion among the Delphi panelists, particularly in reference to those ingredients for which there was notable disagreement regarding their usefulness. Thereafter, the steering committee made minor clarifying edits to the ingredient list and Delphi survey based on panelist feedback. In the second Delphi round, panelists were again asked to rate all ingredients from 1 to 9, while also considering the results of the first survey round and the associated discussion. An ingredient was included in the final recommended list for a particular cosmetic concern if at least 70% of Delphi panelists rated it between 7 and 9 and no more than 15% of participants rated it between 1 and 3 during the second round, and if it was confirmed as acceptable at the consensus meeting.

Ethical considerations

Because this study only included a Delphi survey of expert cosmetic dermatologists, and because any disclosure of human subjects' responses would not reasonably place subjects at risk of criminal or civil liability or be damaging to the subjects' financial standing, employability, educational advancement, or reputation, this study did not meet the criteria for human participant research and did not require institutional review board approval or informed consent, in accordance with 45 code of federal regulations §46. As such, the study was deemed exempt by the Northwestern University Institutional Review Board.

Data collection and statistical analysis

Delphi surveys were administered via a web-based survey software (Qualtrics). Descriptive statistics were provided, and mean scores were computed using Qualtrics and Microsoft Excel version 16.86.

Analysis of published data supporting consensus ingredients

A second literature review was performed following the Delphi process to summarize published data supporting the efficacy of consensus ingredients and corresponding indications. The level of evidence for each was determined according to standards set forth by the Oxford Centre for Evidence-Based Medicine.³

RESULTS

An initial list of 318 cosmetic ingredients obtained from the literature review was reviewed by a panel comprised of 17[†] experts in cosmetic dermatology. A total of 83 ingredients were recommended as useful for at least one of the 7 cosmetic concerns by 2 or more of these panelists, and were thus included in the Delphi consensus process conducted with the second expert panel. The Delphi process occurred from September 2023 to September 2024. Sixty-two panelists (Association of Academic Cosmetic Dermatology members and residency cosmetic directors) across 43 unique institutions completed 2 rounds of Delphi. Following the first round of Delphi, panelists recommended inclusion of 6 additional ingredients, for a total of 89, which were considered in the second Delphi round (Table I).

A total of 23 unique ingredients across 7 skin concerns met inclusion criteria following the second Delphi round and were accepted during the consensus meeting (Table II): ammonium lactate (79% consensus for dry skin); azelaic acid (acne, 87.1%; dark spots, 88.7%); benzoyl peroxide (acne; 95.2%; oily skin, 80.7%); brimonidine (redness, 75.8%); ceramides (dry skin, 82.1%); chemical sunscreen (fine lines and wrinkles, 82.3%); clindamycin (acne, 90.3%); glycolic acid (acne, 79%; dark spots 91.9%); green-tinted products (redness, 77.4%); hyaluronic acid (dry skin, 79%); hydroquinone (dark spots, 98.4%); ivermectin (redness, 72.6%); kojic acid (dark spots, 93.6%); metronidazole (redness, 79%); mineral sunscreen (fine lines and wrinkles, 96.8; redness, 95.2%); niacinamide (redness, 72.6%; dark spots, 79%); petrolatum (dry skin, 85.5%); retinoids (fine lines and wrinkles, 96.8%; acne, 96.8%; dark spots, 96.8%; large pores, 93.6%; and oily skin, 93.6%); salicylic acid (acne, 93.6%, oily skin, 79%);

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Table I. Results of the Delphi consensus process

Skin concern	Ingredient	First round (N = 62)		Second round (N = 62)		Consensus achieved
		Mean score	% of responses 7-9	Mean score	% of responses 7-9	
Fine lines and wrinkles	Ceramides	6.8	54.8	6.7	61.3	No
	Chemical sunscreen	7.1	74.2	6.7	82.3	Yes
	Mineral sunscreen	8.7	95.2	8.8	96.8	Yes
	DNA repair enzymes	5.9	30.6	5.5	17.7	No
	Glycolic acid	6.9	61.3	7.0	66.1	No
	Growth factors	6.4	45.2	6.1	40.3	No
	Peptides	6.5	46.8	6.4	46.8	No
	Hyaluronic acid	6.5	48.4	6.6	48.4	No
	Lactic acid	6.3	40.3	6.0	27.4	No
	Retinoids	8.6	93.5	8.8	96.8	Yes
Acne	Niacinamide (B3)	6.8	58.1	6.8	66.1	No
	Vitamin C	7.9	88.7	7.8	88.7	Yes
	Retinoids	8.8	96.8	8.8	96.8	Yes
	Azelaic acid	7.6	82.3	7.7	87.1	Yes
	Glycolic acid	7.1	69.4	7.1	79.0	Yes
	Salicylic acid	7.8	85.5	7.8	93.6	Yes
	Lactic acid	6.2	46.8	6.2	43.6	No
	Mandelic acid	5.6	24.2	5.4	16.1	No
	Polyhydroxy acid	5.7	25.8	5.5	22.6	No
	Benzoyl peroxide	8.5	95.2	8.4	95.2	Yes
Redness	Clindamycin	7.7	88.7	7.5	90.3	Yes
	Dapsone	6.8	61.3	6.7	67.7	No
	Erythromycin	5.2	27.6	4.9	29.0	No
	Minocycline	6.1	50.0	5.8	51.6	No
	Clascoterone	-	-	5.9	51.3	No
	Ethoxybenzaldehyde	5.0	1.6	5.1	1.6	No
	Aloe vera	4.8	1.6	4.7	0.0	No
	Azelaic acid	6.8	61.3	6.8	62.9	No
	Colloidal oatmeal	5.6	17.7	5.5	16.1	No
	Cucumber extract	5.1	6.5	5.0	3.2	No
	Green tea	5.5	21.0	5.4	19.4	No
	Green-tinted products	7.3	71.0	7.2	77.4	Yes
	Humectants (eg, panthenol, glycerin)	6.2	37.1	6.0	30.7	No
	Hyaluronic acid	6.1	40.3	6.1	37.1	No
	Hydrocortisone	4.0	9.7	3.2	6.5	No
	Ivermectin	7.1	71.0	7.0	72.6	Yes
	Niacinamide (B3)	7.0	62.9	7.0	72.6	Yes
	Oxymetazoline/Brimonidine	7.3	77.4	7.1	75.8	Yes
	Sulfacetamide-sulfur	7.3	75.8	7.2	82.3	Yes
	Chemical sunscreen	6.5	61.3	6.0	59.7	No
	Mineral sunscreen	8.6	95.2	8.6	95.2	Yes
	Metronidazole	7.2	71.0	7.2	79.0	Yes

Continued

Table I. Cont'd

Skin concern	Ingredient	First round (N = 62)		Second round (N = 62)		Consensus achieved
		Mean score	% of responses 7-9	Mean score	% of responses 7-9	
Dark spots	Alpha-arbutin	6.0	33.9	5.8	24.2	No
	Cysteamine hydrochloride	6.4	48.4	6.1	41.9	No
	Hydroquinone	8.4	96.8	8.4	98.4	Yes
	Kojic acid	8.0	90.3	8.0	93.6	Yes
	Niacinamide	7.2	74.2	7.1	79.0	Yes
	Tranexamic acid	7.8	80.6	7.9	87.1	Yes
	Vitamin C	8.0	83.9	8.0	87.1	Yes
	Retinoids	8.4	95.2	8.5	96.8	Yes
	Azelaic acid	7.8	83.9	7.8	88.7	Yes
	Glycolic acid	7.7	85.5	7.8	91.9	Yes
	Salicylic acid	6.8	59.7	6.9	67.7	No
	Corticosteroids alone (eg, fluocinolone acetonide)	5.7	33.9	4.5	21.0	No
	Corticosteroids in combination with other ingredients	-	-	6.4	61.5	No
Large pores	Silicone-based primer	5.9	37.1	5.8	35.5	No
	Azelaic acid	6.0	37.1	5.7	29.0	No
	Citric acid	5.2	16.1	5.2	12.9	No
	Fruit acid	5.4	17.7	5.3	11.3	No
	Glycolic acid	6.5	46.8	6.5	46.8	No
	Lactic acid	6.1	35.5	6.0	29.0	No
	Malic acid	5.6	24.2	5.3	12.9	No
	Salicylic acid	6.8	58.1	6.8	61.3	No
Dry skin	Retinoids	8.2	87.1	8.5	93.6	Yes
	Cyclomethicone	5.9	29.0	5.6	19.4	No
	Dimethicone	6.8	59.7	6.6	56.5	No
	Mineral oil	5.8	35.5	5.6	27.4	No
	Paraffin	5.3	19.4	4.9	11.3	No
	Petrolatum	7.9	87.1	7.9	85.5	Yes
	Propylene glycol	5.0	22.6	4.5	9.7	No
	Squalene	6.0	35.5	5.7	21.0	No
	Ammonium lactate	7.1	71.0	7.1	79.0	Yes
	Glycerin (glycerol)	6.6	46.8	6.4	40.3	No
	Hyaluronic acid	7.3	69.4	7.3	79.0	Yes
	Sodium lactate	6.0	32.3	5.7	22.6	No
	Urea	7.2	72.6	7.3	79.0	Yes
	Glycol stearate	5.5	17.7	5.3	8.1	No
	Sunflower seed oil glycerides	5.6	21.0	5.4	11.3	No
	Ceramides	-	-	7.6	82.1	Yes
Oily skin	Niacinamide	5.9	25.8	5.7	19.4	No
	Papaya	5.1	8.1	5.1	4.8	No
	Polymer-absorbing beads	5.3	11.3	5.2	4.8	No
	Salicylic acid	7.2	69.4	7.2	79.0	Yes
	Witch hazel	4.8	17.7	4.6	14.5	No
	Retinoids	8.4	91.9	8.5	93.6	Yes
	Benzoyl peroxide	7.4	69.4	7.4	80.7	Yes
	Glycolic acid	-	-	6.4	64.1	No
	Mandelic acid	-	-	5.5	28.2	No
	Lactic acid	-	-	5.5	23.1	No

Table II. Skincare ingredients that reached consensus for each skin concern

Skin concern	Ingredient	% of dermatologists who recommended this ingredient (N = 62)
Fine lines and wrinkles	Mineral sunscreen	96.8
	Retinoids	96.8
	Vitamin C	88.7
	Chemical sunscreen	82.3
Acne	Retinoids	96.8
	Salicylic acid	93.6
	Benzoyl peroxide	95.2
	Azelaic acid	87.1
	Clindamycin	90.3
	Glycolic acid	79.0
Redness	Mineral sunscreen	95.2
	Sulfacetamide-sulfur	82.3
	Niacinamide (B3)	72.6
	Green-tinted products	77.4
	Metronidazole	79.0
	Brimonidine	75.8
Dark spots	Ivermectin	72.6
	Hydroquinone	98.4
	Kojic acid	93.6
	Retinoids	96.8
	Vitamin C	87.1
	Azelaic acid	88.7
Large pores	Glycolic acid	91.9
	Tranexamic acid	87.1
	Niacinamide (B3)	79.0
Dry skin	Retinoids	93.6
	Petrolatum	85.5
	Ammonium lactate	79.0
	Hyaluronic acid	79.0
	Urea	79.0
	Ceramides	82.1
Oily skin	Retinoids	93.6
	Benzoyl peroxide	80.7
	Salicylic acid	79.0

sulfacetamide-sulfur (redness, 82.3%); tranexamic acid (dark spots, 87.1%); urea (dry skin, 79%); vitamin C (fine lines and wrinkles, 88.7%; dark spots, 87.1%). Over 95% of panelists recommended mineral sunscreen for fine lines and wrinkles, and redness; retinoids for fine lines and wrinkles, acne, and dark spots; benzoyl peroxide for acne; and hydroquinone for dark spots.

Many consensus ingredients had reasonably strong evidence to support their use for each indication. Forty-four percent were supported by level 1b evidence, 47% by level 2b evidence, and a small minority (5.88%) with no published evidence

for the listed indication. In particular, benzoyl peroxide was unsupported only for oily skin and green-tinted products for mitigation of redness (Table III).

DISCUSSION

Based on a qualitative research approach including a literature search, expert review, and Delphi process, consensus was achieved among U.S.-based cosmetic dermatology experts regarding the skin care ingredients most recommended for common skin concerns.

In general, ingredients that were most recommended tended to be those for which safety and effectiveness were supported by high-level evidence. For instance, retinoids, which were recommended for the treatment of fine lines and wrinkles, acne, and dark spots are probably the most studied topical cosmetic formulation and the one for which there is the highest quality and most longitudinal data (level 1b evidence for each of these 3 indications).^{94,95}

Retinoids were recommended for the treatment of 5 of the 7 cosmetic concerns reviewed, including fine lines and wrinkles, acne, dark spots, large pores and oily skin. Since Kligman et al,⁹⁶ first demonstrated clinical improvement in photodamaged skin with the use of tretinoin 40 years ago, numerous additional studies have confirmed additional retinoid-associated beneficial effects on skin.⁶⁷⁻⁷⁰ Retinoids may improve skin through their anti-inflammatory properties, as well as their ability to stimulate collagen synthesis, increase cell turnover, and reduce epidermal melanin.⁹⁷

Similarly, it is not surprising that mineral sunscreen was recommended for fine lines and wrinkles given the well-accepted finding that protection from ultraviolet radiation can help prevent the physical appearance of premature skin aging (level 1b evidence).²⁶⁻²⁸ Studies have shown consistent sunscreen use reduces skin aging both clinically and histologically.^{26,27} Other recommended ingredients, such as vitamin C and chemical sunscreens, may also slow skin aging by mitigating the effects of ultraviolet light (1b).^{26-28,91-93}

Mineral sunscreen was also recommended for reducing redness, with this benefit possibly deriving from its effect in protecting sensitive skin from external insults and irritation (2b).⁵⁵⁻⁵⁷ The anti-inflammatory ingredients niacinamide and sulfacetamide-sulfur were similarly recommended, as were green-tinted products, which help to camouflage redness.^{37,58-60,80-82}

Benzoyl peroxide and salicylic acid, which were recommended for acne and oily skin, are known

Table III. Summary of the published clinical studies for each consensus ingredient and corresponding indication (for ingredients with multiple published studies available, the publication[s] with the highest level of evidence were included)

Ingredient	Indication	Published supporting clinical studies	Level of evidence
Ammonium lactate	Dry skin	Topical ammonium lactate has demonstrated efficacy in treating dry skin (xerosis), comparable to topical urea and SA. Many studies have focused on foot xerosis: • A systematic review of moisturizers for foot xerosis highlighted that formulations containing ammonium lactate are effective in improving skin hydration and reducing dryness.⁴ • A single-blind RCT demonstrated that 91% of participants using ammonium lactate 12% experienced $\geq 50\%$ improvement in heel dryness after 8 wk, compared to untreated participants who showed no change or worsening.⁵ • Additional double-blinded RCTs found no difference between ammonium lactate 12% vs pure lanolin and SA/urea ointment, respectively.^{6,7}	1b
Azelaic acid	Acne	• Multiple studies have shown that 20% azelaic acid cream effectively reduces both inflammatory and non-inflammatory lesions. Efficacy is comparable to tretinoin 0.05% cream, benzoyl peroxide 5% gel, and erythromycin 2% ointment.⁸⁻¹¹ • Anti-tyrosinase activity of azelaic acid also makes it helpful as an acne treatment for patients prone to postinflammatory hyperpigmentation.¹²	2b
	Dark spots	• A double-blind RCT found that azelaic acid 20% significantly reduced pigmentary intensity in darker-skinned patients over 24 wk compared to vehicle control.¹³ • Another pilot study with azelaic acid 15% gel showed a 2-point improvement in hyperpigmentation after 16 wk, with 31% achieving complete resolution.¹²	2a
Benzoyl peroxide	Acne	Multiple double-blind RCTs have shown benzoyl peroxide (BPO) significantly reduces inflammatory and noninflammatory lesions: • 2.5% BPO formulation was found equally effective as 5% and 10% concentrations, with fewer side effects.¹⁴ • Combination therapies, such as clindamycin-BPO and adapalene-BPO, consistently outperformed monotherapies and vehicle controls in lesion reduction.¹⁵⁻¹⁷ • BPO 5% showed significant improvements in acne lesions within 4 wk compared to isotretinoin 0.05% and placebo.¹⁸ • BPO plus clindamycin significantly reduced inflammatory and total lesions compared to individual agents or placebo in another double-blind RCT.¹⁹	1b
	Oily skin	While BPO is primarily studied for acne, it may reduce oiliness as a secondary outcome. One double-blind RCT comparing BPO to isotretinoin found that BPO 5% gel significantly reduced acne lesions and improved skin texture, which could be associated with oil control. ¹⁸ However, specific studies on BPO's direct effects on sebum production or oily skin are lacking.	5
Brimonidine	Redness	Brimonidine has demonstrated efficacy in reducing facial redness, particularly in rosacea, through multiple high-quality studies: • A double-blind RCT found that brimonidine 0.5% gel significantly reduced erythema compared to vehicle control within 30 min of application, with effects lasting up to 12 h.²⁰ • Another double-blind RCT confirmed the rapid onset of redness reduction with 0.5% brimonidine gel, showing significant improvements in clinician and patient assessments of redness compared to vehicle.²¹ • A single-blind RCT on adjuvant brimonidine with air cooling after IPL treatment showed sustained erythema reduction compared to air cooling alone.²² • An open-label study over 12 mo reported consistent and durable efficacy of brimonidine 0.5% gel in reducing redness without tachyphylaxis.²³	1b

Continued

Table III. Cont'd

Ingredient	Indication	Published supporting clinical studies	Level of evidence
Ceramides	Dry skin	Ceramides have shown efficacy in improving dry skin through several clinical studies, including: • A double-blind RCT demonstrated that a ceramide complex cream significantly improved skin hydration, reduced transepidermal water loss (TEWL), and increased ceramide levels in the stratum corneum compared to placebo over 4 wk.²⁴ • Another double-blind RCT found that 1% ceramide cream was more effective at reducing TEWL in children with atopic dermatitis than lanolin, urea, or vaseline, demonstrating superior skin barrier improvement after 14 d.²⁵	1b
Chemical sunscreen	Fine lines and wrinkles	Chemical sunscreens have demonstrated efficacy in preventing and reducing fine lines and wrinkles associated with photoaging: • An unblinded RCT showed that daily broad-spectrum sunscreen use significantly reduced skin aging over 4.5 y, with 24% less skin aging in the daily-use group compared to discretionary-use controls.²⁶ • A double-blind RCT over 24 mo found that daily sunscreen application significantly reduced solar elastosis, a key marker of photodamage, compared to vehicle control.²⁷ • A single-ingredient trial of SPF 30 sunscreen applied daily for 52 wk demonstrated significant improvements in skin texture, clarity, and photoaging parameters by wk 12, with continued improvement through wk 52.²⁸ <i>*Note:</i> Currently, there is a lack of direct comparative studies between mineral and chemical sunscreens specifically evaluating their efficacy in reducing fine lines, wrinkles, or other signs of photoaging. Both types of sunscreens are effective in protecting the skin from UV radiation, which is a major contributor to skin aging. Tinted sunscreens offer protection against visible light, making them beneficial for individuals prone to hyperpigmentation or visible light–induced photodermatoses (PMID: 32335182).²⁶⁻²⁹	1b
Clindamycin	Acne	• A double-blind RCT comparing clindamycin 1% with benzoyl peroxide 5% or their combination found clindamycin alone significantly reduced inflammatory and total lesion counts compared to placebo.¹⁹ • Another RCT showed clindamycin 1% foam was superior to clindamycin gel and vehicle in reducing total, inflammatory, and noninflammatory lesion counts.³⁰ • A combination gel of clindamycin 1% and benzoyl peroxide 5% outperformed individual agents, significantly reducing lesions and improving global acne severity.¹⁵	1b
Glycolic acid	Acne	• A single-blind split-face trial comparing glycolic acid (GA) to amino fruit acid found that GA significantly reduced inflammatory and noninflammatory acne lesions within the first month of treatment.³¹ • A randomized trial in Indian patients compared glycolic acid peels with salicylic-mandelic acid peels and found both effective, though salicylic-mandelic acid peels were superior for active acne.³² • A split-face trial of 70% glycolic acid vs Jessner's solution showed significant improvement in acne grading for both treatments, with GA being better tolerated.³³	2b
	Dark spots	• A single-blind split-face trial comparing GA to amino fruit acid for melasma showed that both reduced the Melasma Area and Severity Index score, with no significant differences between treatments.³⁴ • A split-face trial of 70% GA versus nanosome vitamin C found that both improved melasma, with vitamin C producing slightly better results.³⁵ • A randomized trial comparing 20% to 35% GA with 10% to 20% trichloroacetic acid for melasma found significant pigment reduction in both groups, with no significant differences in efficacy.³⁶	2b
Green-tinted products	Redness	The evidence supporting green-tinted products for reducing redness is limited and primarily anecdotal. A review article on rosacea management mentioned green-tinted cosmetics as a potential option for masking redness, but no clinical trials have evaluated their efficacy directly. ³⁷	5

Hyaluronic acid	Dry skin	HA has demonstrated efficacy in improving dry skin in multiple clinical studies, including: • A single-blind RCT found that 0.1% HA at different molecular weights significantly improved skin hydration and elasticity compared to placebo.³⁸ • A double-blind RCT comparing low- and high-molecular-weight HA showed that low-molecular-weight HA was most effective in enhancing skin hydration in patients with xerosis cutis.³⁹ • A split-body RCT found that HA foam improved eczema severity and xerosis significantly more quickly than ceramide cream in atopic dermatitis.⁴⁰ • An uncontrolled study of nano-HA for periorbital skin reported a significant increase in skin hydration of up to 96%.⁴¹	1b
Hydroquinone	Dark spots	HQ is well-supported for treating dark spots and hyperpigmentation. Combination topical may be more effective than HQ alone. Potential for side effects may make alternative lightening agents preferable: • A systematic review and meta-analysis found that HQ monotherapy and combination therapy demonstrated comparable efficacy to TXA, azelaic acid, and kojic acid. HQ combination therapy conferred the highest risk of irritation.⁴² • A systematic review found that HQ 2% and 4% were effective in reducing postinflammatory hyperpigmentation. Combination topical may be more effective than HQ alone.⁴³ • A double-blind RCT of 4% HQ showed complete clearance of melasma in 40% of patients compared to 10% in the placebo group.⁴⁴ • A split-face RCT found 4% HQ improved melasma in 76.9% of patients, significantly outperforming placebo.⁴⁵ • A comparative RCT of 4% HQ and a retinol-free brightener showed both treatments improved melasma severity, with HQ achieving greater reductions in hyperpigmentation.⁴⁶	1b
Ivermectin	Redness	Ivermectin has demonstrated efficacy in reducing redness associated with papulopustular rosacea: • A double-blind RCT found 1% ivermectin cream significantly reduced erythema and achieved higher treatment success compared to vehicle control by wk 4, with continued improvement through wk 12.⁴⁷ • A single-ingredient trial reported a complete response in 31.6% of patients at wk 8, with progressive erythema reduction over 20 wk.⁴⁸	2b
Kojic acid	Dark spots	• A double-blind RCT comparing kojic acid with 4% HQ found kojic acid provided a higher rate of fading in skin hyperpigmentation.⁴⁹ • A split-face RCT with a kojic acid gel containing 10% glycolic acid and 2% HQ showed greater melasma clearance on the kojic acid side compared to the control.⁵⁰ • A single-blind RCT found that combining kojic acid with HQ or corticosteroids improved melasma more effectively than kojic acid alone.⁵¹	2b
Metronidazole	Redness	Topical metronidazole has demonstrated efficacy in reducing redness, particularly in rosacea: • A double-blind RCT comparing 1% metronidazole cream with SPF 15 to vehicle showed significant improvements in erythema, inflammatory lesion count, and global assessment scores.⁵² • A single-blind RCT comparing 0.75% and 1% metronidazole cream found both concentrations reduced erythema, with 1% being slightly more effective.⁵³ • An open-label phase 4 trial of 0.75% metronidazole gel demonstrated significant reductions in erythema severity, with nearly 50% improvement by wk 12.⁵⁴	2b
Mineral sunscreen	Fine lines and wrinkles	See chemical sunscreen.	1b

Continued

Table III. Cont'd

Ingredient	Indication	Published supporting clinical studies	Level of evidence
Niacinamide	Redness	Sunscreen has been shown to provide protection against erythema caused by UV exposure. Published studies do not specify whether chemical or mineral sunscreen was used: • A split-face RCT comparing SPF 50 to SPF 100 sunscreen found SPF 100 reduced erythema intensity more significantly after sun exposure.⁵⁵ • An RCT of sunscreen vs vehicle control found that erythema developed on untreated skin but not on sunscreen-protected sites after sub-erythral UV exposure.⁵⁶ • An unblinded RCT comparing instructed SPF 15 use to typical sunscreen application during sunny holidays showed virtually no erythema in the instructed group, highlighting sunscreen's protective effect.⁵⁷	2b
	Redness	Niacinamide has shown promise in reducing redness, particularly in inflammatory skin conditions: • An open-label RCT comparing 4% niacinamide to placebo for seborrheic dermatitis demonstrated a 75% reduction in redness and scaling with niacinamide, compared to 35% with placebo.⁵⁸ • A double-blind RCT evaluating 2% and 4% niacinamide for discoid lupus erythematosus showed significant reductions in erythema, with better results for the 4% formulation.⁵⁹ • A double-blind RCT comparing 4% niacinamide to ketoconazole for seborrheic dermatitis found that niacinamide reduced erythema more effectively than ketoconazole.⁶⁰	2b
	Dark spots	Niacinamide has demonstrated efficacy in improving dark spots and hyperpigmentation: • A double-blind split-body RCT comparing 4% niacinamide to 4% HQ for melasma found both treatments improved pigmentation, with no significant difference between them, and 44% of patients reporting improvement with niacinamide.⁶¹ • A single-blind split-face RCT using niacinamide in a cationic liposome formulation showed significant improvements in skin brightness, melanin index, and melasma area compared to placebo.⁶² • A double-blind RCT evaluating 4% niacinamide for axillary hyperpigmentation showed a good to excellent response in 24% of cases, compared to 6% with placebo.⁶³	2b
Petrolatum	Dry skin	Petrolatum is effective for dry skin, improving barrier recovery and hydration, though evidence is primarily based on small, observational trials. • An unblinded split-arm RCT demonstrated that petrolatum significantly accelerated skin barrier recovery and reduced trans-epidermal water loss (TEWL) after acetone exposure.⁶⁴ • A split-face trial comparing white petrolatum to a newer petrolatum-based formulation (balm) found both significantly reduced dryness and TEWL while improving stratum corneum hydration, with the newer formula being more tolerable.⁶⁵ • An open-label study found that petrolatum significantly reduced inflammatory cytokines in xerotic skin, supporting its role in improving hydration and reducing inflammation.⁶⁶	2b
Retinoids	Fine lines and wrinkles	Retinoids are well-supported for improving fine lines and wrinkles: • A double-blind RCT of 0.1% tretinoin cream vs placebo showed significant improvements in fine wrinkling, mottled hyperpigmentation, and overall photodamage after 24 wk.⁶⁷ • A double-blind RCT of 0.05% tretinoin cream demonstrated greater improvement in clinical signs of photodamage (fine and coarse wrinkling, pigmentation) compared to placebo.⁶⁸ • A double-blind RCT comparing 0.1% tazarotene to 0.05% tretinoin found both reduced fine wrinkles, with tazarotene demonstrating superior results in certain efficacy measures.⁶⁹ • A double-blind RCT of retinol vs tretinoin for photoaging found both significantly improved wrinkles and global photodamage, with no statistically significant differences between them.⁷⁰	1b

Acne	Retinoids are effective for acne, particularly for reducing both inflammatory and non-inflammatory lesions, and are supported by robust clinical evidence: • A double-blind RCT of 0.1% adapalene lotion vs vehicle showed significant improvement in acne lesions (total, inflammatory, and non-inflammatory) and increased Investigator Global Assessment success rates.⁷¹ • A 52-wk single-ingredient trial of 0.3% adapalene gel demonstrated >75% reduction in total, inflammatory, and non-inflammatory lesions by the end of the study.⁷² • A randomized trial comparing adapalene gel to adapalene microspheres showed both formulations significantly reduced lesion counts, with microspheres being less irritative.⁷³ • A double-blind RCT of 0.1% tretinoin gel showed statistically significant reductions in acne severity and lesion counts compared to placebo.⁷¹	1b
Dark spots	Retinoids are effective for reducing dark spots and improving overall skin pigmentation. Studies focus primarily on photodamage, with fewer studies directly targeting melasma or other specific hyperpigmentation conditions: • A double-blind RCT of 0.1% tretinoin gel showed significant improvements in mottled hyperpigmentation and overall photodamage compared to placebo after 24 wk.⁶⁷ • A double-blind RCT of 0.05% tretinoin cream for photodamage demonstrated greater reductions in pigmentation and lentigines compared to placebo.⁶⁸ • A randomized split-face trial comparing 0.3% adapalene to 0.05% tretinoin for photoaging found both significantly reduced pigmentation and melasma severity, with no significant differences in efficacy.⁷⁴ • A double-blind RCT comparing 0.1% tazarotene to 0.05% tretinoin showed both significantly reduced hyperpigmentation, with tazarotene having slightly better outcomes in certain measures.⁶⁹	1b
Large pores	Retinoids have demonstrated potential in improving large pores, though evidence is more limited compared to other indications: • A single-blind RCT comparing daily tazarotene (0.01% to 0.1%) and 0.05% tretinoin found both treatments reduced pore size and improved skin texture, with tazarotene showing slightly greater efficacy.⁷⁵ • A single-ingredient trial evaluating low- and high-concentration retinol (1500-6600 IU) found that higher concentrations significantly reduced pore size and improved skin texture compared to lower concentrations.⁷⁶	2b
Oily skin	Retinoids have limited but promising evidence for reducing sebum production and improving oily skin: • A randomized trial comparing adapalene gel to other topical agents showed a reduction in sebum production as a secondary outcome.⁷³	2b
Salicylic acid	Acne SA is effective for reducing acne lesions, especially comedonal acne, with evidence supporting its safety and efficacy. Studies vary in concentrations and formulations used, limiting generalizability: • A single-ingredient trial of 30% supramolecular SA peels used biweekly for 2 mo showed significant reductions in acne severity (Global Acne Grading System score) and improvements in the skin microenvironment.⁷⁷ • An unblinded RCT comparing Jessner's solution to 30% SA peels found that SA peels were more effective in treating noninflammatory lesions and overall acne improvement.⁷⁸ • A review of 4 clinical studies highlighted the efficacy and safety of 0.5% to 2% SA solutions, demonstrating reductions in primary lesions and overall acne severity.⁷⁹	2b
	Oily skin SA shows promise for reducing oiliness, likely through its ability to penetrate pores and modulate sebaceous activity, though further dedicated studies are needed. • A single-ingredient trial of 30% supramolecular SA peels applied biweekly for 2 mo showed significant reductions in sebum levels and an association between reduced oiliness and acne improvement (PMID: 36350527).⁷⁷ • An unblinded RCT comparing SA peels with Jessner's solution found that SA reduced sebaceous activity alongside its efficacy in treating acne.⁷⁸	2b

Continued

Table III. Cont'd

Ingredient	Indication	Published supporting clinical studies	Level of evidence
Sulfacetamide-sulfur	Redness	Sulfacetamide-sulfur is effective for reducing redness in rosacea, particularly when combined with sunscreen, though further well-designed studies are needed. • An open-label RCT comparing sulfacetamide-sulfur (10%/5%) to vehicle showed significant reductions in facial erythema, with 83% of patients improving by wk 8 compared to 31% in the control group.⁸⁰ • A single-blind randomized study comparing sulfacetamide-sulfur with sunscreen to metronidazole cream for rosacea found greater improvements in inflammatory lesions and erythema with sulfacetamide-sulfur.⁸¹ • A nonrandomized open-label series demonstrated clinically significant reductions in erythema and an improved quality of life in patients with papulopustular rosacea treated with sulfacetamide-sulfur.⁸²	2b
Tranexamic acid	Dark spots	TXA is effective for reducing dark spots, with results comparable to HQ monotherapy and combination therapy. TXA appears to be less irritating than other topical lightening agents, including HQ: • A systematic review and meta-analysis concluded that topical TXA significantly improved melasma, particularly in reducing the Melasma Area and Severity Index and was well-tolerated across studies.⁸³ • Another systematic review and meta-analysis found that topical TXA was comparable in efficacy to HQ monotherapy and combination therapy, azelaic acid, and kojic acid. Of these, topical TXA also had the lowest risk of irritation.⁴² • A double-blind split-face RCT comparing 5% topical TXA to 4% HQ found both treatments significantly reduced melasma severity, with TXA showing comparable results to HQ.⁸⁴ • An unblinded split-face RCT of topical TXA combined with IPL vs IPL alone demonstrated greater reductions in melanin index and melasma severity on the TXA-treated side.⁸⁵ • A double-blind RCT comparing 5% topical TXA to 2% HQ showed significant reductions in melasma severity in both groups, with no statistically significant differences between them.⁸⁶ • A single-ingredient trial of topical TXA serum found 79% of patients noticed improvements in skin tone and dark spots after 2 wk.⁸⁷	1b
Urea	Dry skin	Topical urea is effective for treating dry skin, particularly in conditions like xerosis and ichthyosis vulgaris, with consistent evidence supporting its role in improving skin hydration and barrier function. • A double-blind RCT in diabetic patients with foot xerosis found 10% urea cream significantly improved fissure healing and xerosis after 4 wk compared to placebo.⁸⁸ • A single-blind split-face RCT comparing 10% urea cream to a glycerol-based emollient for ichthyosis vulgaris showed superior efficacy of urea in reducing scaling, roughness, and redness.⁸⁹ • A pilot registry study in diabetic patients with foot xerosis found 5% urea cream improved skin hydration, increased skin thickness, and reduced transepidermal water loss compared to untreated controls.⁹⁰	2b
Vitamin C	Fine lines and wrinkles	Topical vitamin C has shown efficacy in reducing fine lines and wrinkles, primarily through its antioxidant properties and role in collagen synthesis: • A meta-analysis reviewing clinical trials confirmed the efficacy of topical vitamin C in reducing UV-induced skin aging, including fine lines, with effects being dose-dependent.⁹¹ • A double-blind split-face RCT of topical vitamin C (10%) over 12 wk demonstrated significant reductions in fine lines and improvement in skin texture on photodamaged skin compared to placebo.⁹² • A double-blind RCT with 5% vitamin C cream applied daily for 3 mo showed improvements in fine wrinkles and overall skin texture, supported by histologic evidence of enhanced collagen production.⁹³	1b

Dark spots	Topical vitamin C is effective for reducing dark spots and improving hyperpigmentation, supported by evidence of its ability to inhibit melanin synthesis and repair UV-induced pigmentation: ⁹²	1b
	• A double-blind split-face RCT of 10% vitamin C showed significant improvement in hyperpigmentation, including melasma, after 12 wk of daily application compared to placebo. • A single-blind split-face trial comparing 70% glycolic acid to nanosome vitamin C with iontophoresis in melasma found both treatments effective, with vitamin C producing better results in reducing pigmentation.³⁵ • A meta-analysis of 31 clinical trials found that vitamin C significantly reduced UV-induced pigmentation in a dose-dependent manner, improving overall skin brightness.⁹¹	

BPO, Benzoyl peroxide; GA, glycolic acid; HA, hyaluronic acid; HQ, hydroquinone; IPL, intense pulsed light; RCT, randomized controlled trial; SA, salicylic acid; SPF, sun protection factor; TEWL, transepidermal water loss; TXA, topical tranexamic acid; UV, ultraviolet.

for their antibacterial and comedolytic properties (benzoyl peroxide: 1b and 5 for acne and oily skin, respectively; salicylic acid: 2b for both).^{14-19,77-79} Additionally, benzoyl peroxide prevents bacterial resistance and has been shown to increase the efficacy of topical antibiotics when compared to topical antibiotics alone.⁹⁸

For the treatment of dry skin, dermatologists recommended ingredients that help to provide hydration and maintain the skin barrier. These included petrolatum (2b),⁶⁴⁻⁶⁶ ammonium lactate (1b),⁴⁻⁷ hyaluronic acid (1b),³⁸⁻⁴¹ urea (2b),⁸⁸⁻⁹⁰ and ceramides (1b).^{24,25}

Several ingredients which are commonly used in cosmetic products were not recommended by the expert panel. This may have been because experts were uncertain regarding the effectiveness of these ingredients for addressing specific cosmetic dermatologic concerns. For instance, while ingredients containing DNA repair enzymes, growth factors, and peptides, are ubiquitous in over-the-counter formulations, the studies supporting their utility for common cosmetic indications may be perceived as insufficient or preliminary. For the treatment of acne, erythromycin, minocycline, dapsone, and clascoterone did not reach consensus. Although many of these products are commonly prescribed, and known to be effective, some panelists noted in their comments that cost and insurance coverage played a role in their utilization. For example, when treating acne, topicals such as mandelic acid and polyhydroxyl acid are more expensive and less likely to be covered by insurance than vitamin A derivatives. Additionally, most of the conditions addressed in this study are considered aesthetic and therefore associated treatments are generally not covered by payers, so the out-of-pocket cost is important to consider. Oral antimicrobials may also have been nonpreferred given the risk of drug resistance and adverse events from extended use.

Limitations of this study included reliance on an expert panel comprised of practicing cosmetic dermatologists who were also professors of dermatology, researchers, and teachers of cosmetic dermatology. While this group could be said to be uniquely qualified to make the requested determinations, it is possible that other cosmetic dermatologists may feel differently. Additionally, all results were based on expert opinion. Although a literature review was used to develop a long list of topical cosmetic ingredients, an evidence-based systematic review or meta-analysis was not conducted. These were deferred in large part due to the heterogeneity of the data, the relative lack of high-level evidence, and

concerns about evidentiary bias associated with studies on proprietary dermatologic formulations.

CONCLUSION

Given the proliferation of novel and reformulated cosmetic ingredients, and the relentless advertising and marketing campaigns touting the benefits of the products that contain them, selecting among such products and deciding which ingredients to prioritize can be a bewildering task even for well-informed patients and their doctors. In this study, a qualitative research approach based on a literature review, expert assessment, and a formal consensus process among practitioners and teachers of cosmetic dermatology was used to develop specific recommendations to improve patient education and provide guidance regarding the selection of cosmetic ingredients. Twenty-three topical cosmetic ingredients were recommended for 7 common dermatologic concerns. Topical retinoids and sunscreens were deemed particularly useful. Patients and their dermatologists can use this information as a starting point for building a skincare routine.

Conflicts of interest

Dr Cohen is or has been a consultant or clinical trial participant in the past 2 years for Revision, Sente, Biopelle, Alastin, Elta, PCA, Allergan, and Merz. Dr Council is a consultant and investigator for Castle Biosciences and a consultant for Sanofi/Regeneron. Dr Goldman owns stock and participates on the Advisory Board of Pomega and Cell Research Corporation; consults, receives honoraria from, and conducts medical research for SkinCeuticals, SkinMedica, and Isdin. Dr Hooper is a consultant for and participates on the Scientific Advisory Board for Revision Skincare. Dr Rossi is a consultant for Evolve CME, Almirall, Merz, Dynamed, Canfield Scientific, Evolus, Biofrontera, QuantiaMD, Lam Therapeutics, Regeneron, and Cutera; has traveled with Mavig and L'Oréal; is on the advisory board for Allergan and Skinfix; is founder of DAR Companies. Drs Alvarez, Kang, Richmond, Hoss, Sulewski, Minkis, Rozenberg, Antonovich, Boucher, Bernstein, Bertucci, Chapas, Dover, Geronemus, Given, Goldbach, Kaufman, Munavalli, Pacheco, Wilson, and Alam have no conflicts of interest to declare.

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